Agent harness: the decision for DCLab

Conclusion of two research reports: *Agent harness choice for DCLab* (2026-10-22) and *Self improving agent harnesses* (2026-10-23). 23 sources of code, papers and docs; DCLab's own code audited at commit aee3f458.

The answer in one paragraph

Keep DCLab's own harness. Do not switch to a framework, and do not adopt Harness² as code. Fix what the harness *remembers*, then teach it to learn, step by step: Claude writes the lessons, a cheap local Qwen model does the work, and DCLab's own grader decides what stays.

Think of the harness as a kitchen and the model as the chef. DCLab's kitchen already has a strict door guard (every action is checked), a rulebook (the workflow validator can say no) and a till (spending stops before it passes a cap). No kit on the market has all three. What the kitchen lacks is a **good camera**: today it records only the first 360 characters of what each tool returns. That is why one training set built from those records had only **4 rows, 1 of them clean**.

Decisions to make

Tick one box per row. The first column is the recommendation from the evidence.

#	Decision	Recommended	Effort / cost	Your choice
D1	Keep the in-house harness; no switch to LangGraph, CrewAI, OpenAI/Google/Microsoft/Claude SDKs	**Yes**	none	[ ] Yes [ ] No [ ] Later
D2	Fix the memory: one whole record per step; table cells masked, not whole exchanges dropped; an export to the training format	**Yes, first**	about 1–2 weeks, €0	[ ] Yes [ ] No [ ] Later
D3	Tests can never reach a live model unless a test asks for one	**Yes**	a few days, €0	[ ] Yes [ ] No [ ] Later
D4	Case memory and an ACE-style playbook (lessons added, never rewritten, each with helped/hurt counts)	**Yes**	about 2 weeks; cents per run	[ ] Yes [ ] No [ ] Later
D5	Probe gate: a lesson stays only if the grader score rises with it and no false alarm appears; with brakes and a lesson-leak filter	**Yes**	a few days, €0 (Qwen)	[ ] Yes [ ] No [ ] Later
D6	Skills as short procedures + scripts, approved by you	**Yes, after D4**	hours of review per skill	[ ] Yes [ ] No [ ] Later
D7	GEPA now and then, on one role's prompt, dev sets only	**Yes, later**	€5–10 per run	[ ] Yes [ ] No [ ] Later
D8	Fine-tune Qwen on the M1 Max (QLoRA SFT, then DPO on the right-vs-wrong pairs)	**Yes, when D2 is done and the Mac is here**	about 30 h a run, €0	[ ] Yes [ ] No [ ] Later
D9	Reinforcement learning on rented GPUs (Agent Lightning or rLLM, DCLab's grader as reward)	**Later**	rented GPUs; your budget	[ ] Yes [ ] No [ ] Later
D10	Harness² code, Darwin Gödel Machine, ADAS, Mem0/Zep, CrewAI	**No**	—	[ ] Yes [ ] No [ ] Later
D11	Legal check: may Claude's runs train a Qwen model? (Anthropic terms restrict training 'competing' models)	**You decide**	your check	[ ] Yes [ ] No [ ] Later

1. “Our harness is not good enough”: tested against the evidence

Claim	Verdict	Evidence, with a DCLab example
The agent loop is primitive; a framework would do better	**Wrong**	The loop is small (224 lines) and sound. Every write passes a guard; the workflow validator can refuse a move; the gateway refuses a request before it could pass a cap; tests run on scripted models. *Example:* when the intern tries to run the final stage before the solution is signed, the validator blocks it. None of the 11 frameworks studied has this rule built in.
The harness does not give the data we need	**Right**	Traces keep 360 characters of each tool result; the intern's model exchanges are kept as shape only; six places record steps, each cut differently. *Example:* the solver set had 4 training rows on 2026-10-21.
The harness is not safe enough in tests	**Right**	A live key reached tests 4 times. *Example:* on 2026-10-21 a gate let your OpenAI key plan an intern turn (24 tool calls) and spend on stage explanations.
DCLab does not learn from its runs yet	**Right**	Phase 9 is still a plan. *Example:* online_shoppers was missed by the rules and by Claude; nothing remembers that lesson for the next shop table.
Something like Harness² is the way to fix it	**Wrong as code, right as ideas**	Harness² has 3 commits, 0 merged pull requests, needs Linux and Google Vertex AI, has no spend cap, and its editor may delete guardrails. Its useful ideas (test each lesson with and without it; skills over prose) fit inside DCLab's harness.
Self-improvement will make Claude much better	**Wrong**	Strong models gain only 1–3 points; weak models gain a lot (one gap shrank from 8.2 to 1.8 points). The prize is a cheap local Qwen doing work close to Claude's.

2. The options compared, in short

Option	What it is	Fit for DCLab	Why
Claude Agent SDK	Anthropic's agent kit	No	Alpha; runs Claude Code in the background; no test model; Claude only
OpenAI Agents SDK	OpenAI's agent kit	No	Best testing, but reaches Claude only through beta adapters; traces go to OpenAI by default
Google ADK	Google's agent kit	No	Native Claude, but its approval pause is experimental; changed its default model silently
Microsoft Agent Framework	AutoGen + Semantic Kernel	Maybe later	Good telemetry; its Claude and Ollama connectors are still beta
LangGraph	graph + checkpoints	Keep, thinly	Already used for durable pauses in the orchestra layer
Pydantic AI	typed agent loop	Fallback	Best pre-call hooks and test models; changes almost daily
CrewAI / smolagents / Letta / Mastra	agent frameworks	No	Telemetry on by default / stalled / retired in Python / TypeScript only
Harness²	self-improving harness	Ideas only	Unmaintained, Linux + Vertex, no spend cap, can delete guardrails
Meta-Harness, RRSI	offline harness search; brakes	Ideas, later	Copy RRSI's brakes now; a small offline search later
DGM, ADAS, SICA, AFlow	agents that rewrite their own code	No	Costly (one DGM run ~$22,000), old code, cheated its own checks
ACE	evolving playbook	**Yes (rebuild)**	42.4→59.4 on AppWorld; small algorithm; append-only
GEPA	prompt improver	**Yes (library)**	MIT; plugs in with 2 functions; budget cap
mlx-lm (+ mlx-lm-lora)	fine-tuning on Apple silicon	**Yes**	QLoRA SFT and DPO on the M1 Max
Agent Lightning / rLLM	RL around an existing agent	**Later**	No harness rewrite; needs NVIDIA GPUs
inspect-ai	evaluation harness	Small trial	Repeated runs and scorers; can wrap the intern

3. The plan, in order, each step with a real DCLab example

Step 1. Record every step whole (D2, D3) *(1–2 weeks, €0)*

What	Every step keeps the system prompt, the tool list, the model's reasoning, the tool call and the tool's full result. Table cells are replaced by typed placeholders, so privacy holds without throwing the record away. Tests can reach a live model only on purpose.
DCLab example	The fraud solve of 2026-10-21 kept its whole results only because the intern was patched by hand that day; every other agent still cuts at 360 characters.
Evidence	Every later step depends on this.

Step 2. Case memory and a playbook (D4) *(about 2 weeks; cents per run)*

What	After each graded run, a short lesson is added to a playbook for that kind of problem, or to a general one about how the agent works. Lessons are only added, never rewritten; each counts how often it helped and hurt. Past cases are found again by problem kind and data bands.
DCLab example	Lesson from online_shoppers: “If the prediction is made at the start of a visit, columns that count pages or time during the visit are not known yet.”
Evidence	ACE: 42.4→59.4 on AppWorld; a full rewrite once collapsed a playbook from 18,282 to 122 tokens and fell below no learning.

Step 3. The probe gate and the brakes (D5) *(a few days, €0 with Qwen)*

What	A new lesson runs with and without it on 10 practice tables; it stays only if the grader score rises and no false alarm appears. Brakes: a limit on edits per try, no lesson tied to one benchmark, and no lesson that names the table's own columns or answers.
DCLab example	This week my first time-column guard looked safe but lowered leak catches from 2 to 1 of 13. A probe gate would have rejected it automatically.
Evidence	RRSI: +6.0 on the search split, only +4.3 outside it, even with brakes.

Step 4. Skills you approve (D6) *(hours of review per skill)*

What	A repeated lesson becomes a short procedure plus a small script that the agent loads only when needed. You approve each one.
DCLab example	A skill “check the time order”: find a date or a counter that only goes up, and split by it (the fraud Time fix of this week, as a reusable step).
Evidence	In Harness², removing skills cost 5.3 points; removing prompt text cost 1.5.

Step 5. GEPA on one prompt (D7) *(€5–10 per run)*

What	Now and then, GEPA improves one role's prompt by reading the runs where it failed. Scored on dev sets only, never on sealed tests.
DCLab example	Improve the leakage reviewer's prompt with the planted-truth tables as the score, like bank_marketing's duration (a known leak).
Evidence	GEPA beat RL by 6% with up to 35× fewer runs (its authors).

Step 6. Fine-tune Qwen on the M1 Max (D8) *(about 30 hours a run, €0)*

What	Train a small local model on DCLab's own whole records (QLoRA SFT, one example per agent turn), then on right-vs-wrong pairs (DPO). Measure it against the rules, not only against the base model.
DCLab example	The 220 pairs, like “duration is a leak (the truth) vs not a leak (the engine's answer)”, teach the model exactly where the rules were wrong.
Evidence	8B 4-bit SFT runs on Apple silicon; RL there runs at about 0.02 steps a second, so not on the Mac.

Step 7. Later: reinforcement learning on rented GPUs (D9) *(your budget)*

What	Agent Lightning or rLLM trains the model by letting it try tasks, scored by DCLab's own grader, with no harness rewrite.
DCLab example	The model solves 100 practice fraud tables; a solve that catches the planted leak earns the reward.
Evidence	Agent Lightning: 41.8→56.4% on SWE-bench Verified with 6K samples (its authors).

4. Risks, and the code that controls each

Risk	What it looks like in DCLab	Control (code, not a prompt)
Overfitting	A lesson helps the practice tables but not new ones	Sealed variants read once; keep 2–3 playbook versions and promote the one that wins most check tables
Reward hacking	A learner 'learns' to use cancellation_reason, a column filled only after a customer leaves	Grader, answer keys, split and holdout stay outside anything the learner can see or edit
Rules loosened	A change quietly removes a guard to pass a test	Validator, caps, grader and rule files are read-only to learning; a refusal test runs after every candidate
Lesson leak	A lesson carries the answer: “expected: group k-fold on order_id” (in today's Phase 9 plan)	The grader gives only error codes; a filter rejects lessons naming the table's own columns or values
Cost	Claude as the solver costs about €3 per task, €90 for 30 challenges	Qwen solves, Claude only teaches; every learning step goes through the gateway's caps
Privacy	Whole records hold more	Cells replaced by placeholders; consent rules tested before any record leaves a workspace

5. What to measure, so we know it works

Measure	Today	Healthy sign
Clean training rows per run	1 clean row in 4 (solver set, 2026-10-21)	most runs give a whole, graded record
Live vs scripted pass rate (harness suite)	10/17 and 14/17 live; 17/17 scripted	the live gap closes
Live model calls during tests	4 known leaks	zero, checked by the gateway
Sealed score per playbook version, paired, at least 3 repeats	not measured yet	above no-learning, best-of-k and retry-without-memory
Qwen vs Claude gap on the same tables	not measured yet	shrinks with each version
Refusal test (rules still refuse bad moves)	—	100%, every candidate
Lessons naming table columns or answers	—	zero
Cost per passed task	—	falls over time
Solver: validation-to-holdout gap; planted leaks caught	15/16 solver runs pass	gap small and stable; more leaks caught

Honest limits

No system in these reports has been tested on business tables with leakage. Most numbers come from each method's own authors; no independent head-to-head benchmark of the frameworks exists. With 30 sealed tables, one table is worth 3.3 points, so a 2-point gain is invisible without paired, repeated runs. The cost figures above are estimates, not measurements.

Decision

Decided by	Date	Decisions approved (D1–D11)	Notes

Full reports: reports/Agent harness choice for DCLab.md and reports/Self improving agent harnesses.md (each with a Persian summary).